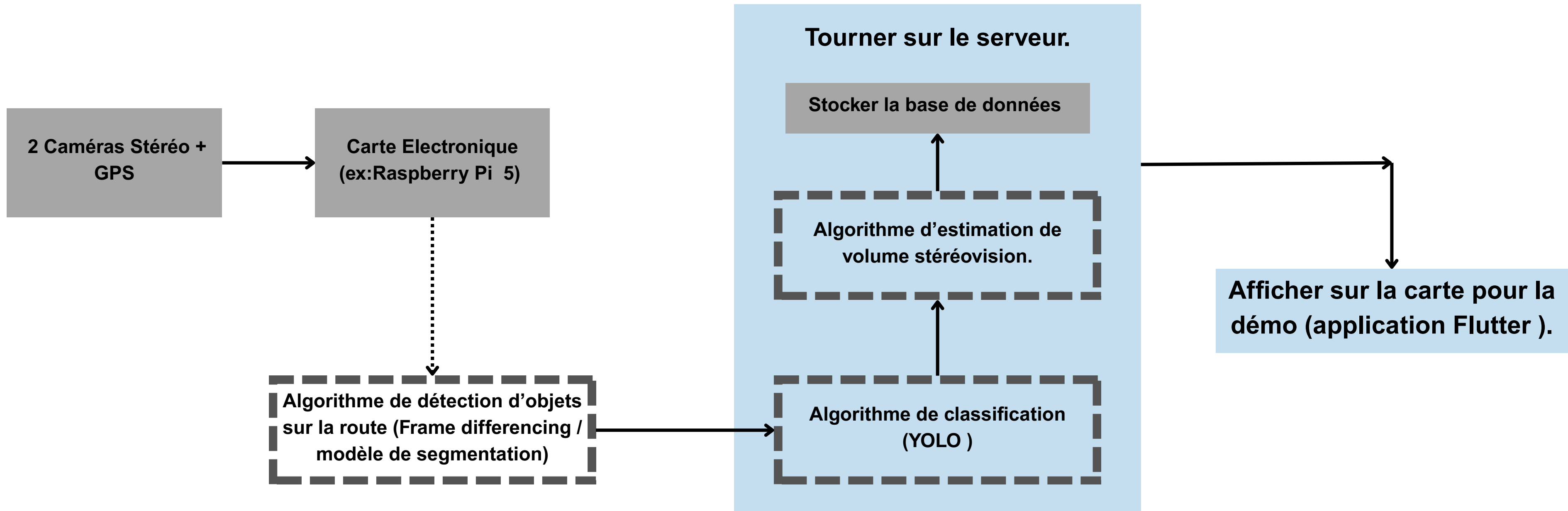


Project Clear Road

Architecture Système



Détection embarqué

Recherche de solutions



Fig. 3. (a) Initial prototype, (b) clustering, (c) filtering.



Fig. 4. (a) Prototype deduced from previous image, (b) clustering, (c) filtering.

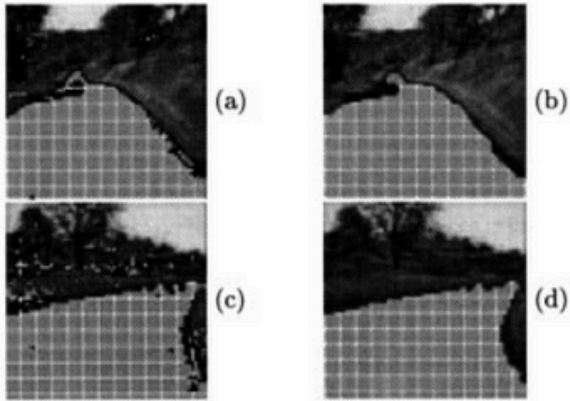


Fig. 5. (a) Clustering, (b) filtering, (c) clustering, (d) filtering.

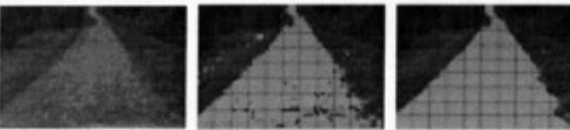


Fig. 6. (a) Forest path image, (b) clustering, (c) filtering.

line, as the percentage of curb points of the total number of line points. A line is considered a curb if it has a score higher than 40%. The best two (or one, or none) curbs are selected, having the highest scores.

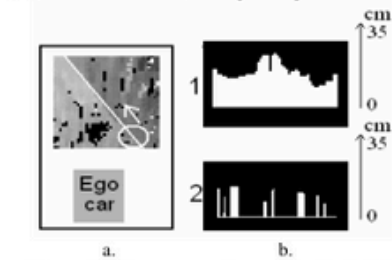


Figure 10. The process of computing the height variation profile of a line is performed by sliding a circular mask in a. In b the profile is drawn for lines marked as 1 and 2 in the Hough space from Figure 7. Only line 1 is classified as curb.

4. Results

The algorithm was implemented in C++. The dense 3D information was generated using a calibrated stereo rig with grayscale cameras and a commercial dense stereo board [10].

Due to the use of software-specific C optimizations and the elevation map representation, a processing time of 10 to 12 ms was achieved for the algorithm itself (on Pentium 4 at 2.6 Ghz). Building the elevation map is the most time consuming part, taking between 6 to 8 ms. The detection of curbs on the elevation map is very fast, about 4 ms.

Overall, with the image acquisition and the dense hardware reconstruction, a sustained processing frame-rate of 35 fps is obtained.

Regarding the robustness of the algorithm we performed the following evaluation that will be extended in the future:

- A number of 50 stereo images of different (random) scenes were selected (out of several hours of stored stereo images, recorded while driving the ego-car).
- Results were analyzed for each frame in terms of missed curbs and false curbs. Visual validation was performed by checking the projection of each 3D curb back onto the image plane.

For curbs higher than 5 cm the algorithm proved very robust, with no missing curbs (80 curbs were

present in the 50 scenes, 3 of the scenes are shown in Figure 13). On one of the 50 scenes, a false curb appeared due to poor 3D reconstruction (Figure 12).

Even though the proposed method was not designed for small curbs, it was also tuned and tested on smaller curbs (3-4 cm) but it is less robust: about 15% of the detected curbs were false ones. This is because small curbs have the height too close to the uncertainty of the stereo reconstruction. Future improvements of the stereo board used will improve the behavior for small curbs.

The algorithm also works for curved curbs, assuming the curvature is low (Figure 13.c).

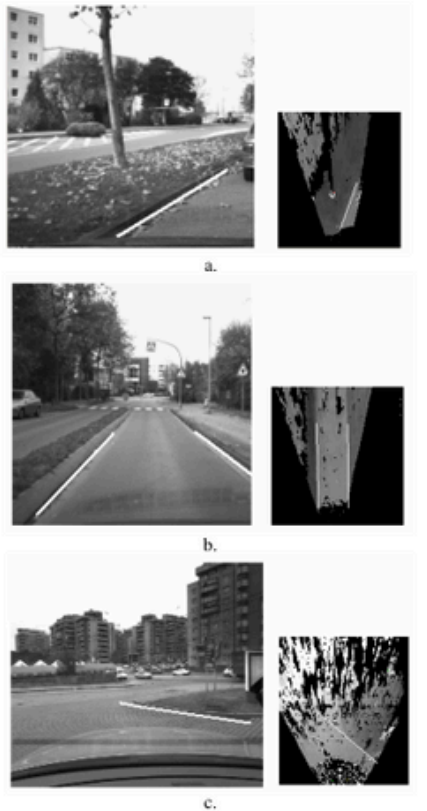


Figure 13. Results for various scenes, curbs are drawn on the elevation map and projected back onto the left image.



INTERNSHIP REPORT

Freespace Detection on Unstructured Roads for Autonomous Driving



GAËLLE DUMONT LE BRAZIDEK
MASTER 2 - ISI

Internship Supervisor: PAVAN VASISITA
Referent Teacher: SYLVAIN ARGENTIERI

February 17th 2025 — August 1st 2025

“A Dynamic Vision Algorithm to Locate a Vehicle on a Nonstructured Road” by R. Aufrère et al.

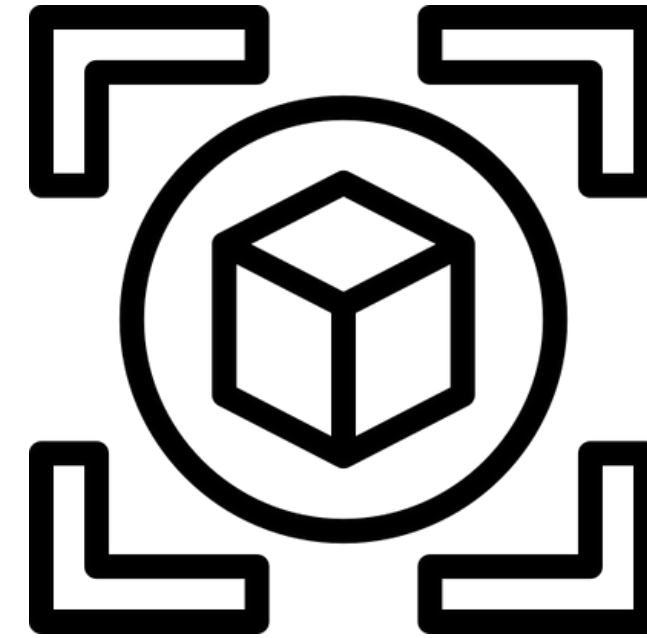
“Curb Detection Based on Elevation Maps from Dense Stereo” by Florin Oniga et al.

“Freespace Detection on Unstructured Roads for Autonomous Driving” by Gaëlle Dumont le brazidek

Solution finale



Road Segmentation



Object Detection



Road Segmentation v1 v2 v3



First iteration



Third iteration

Dataset : lost and found

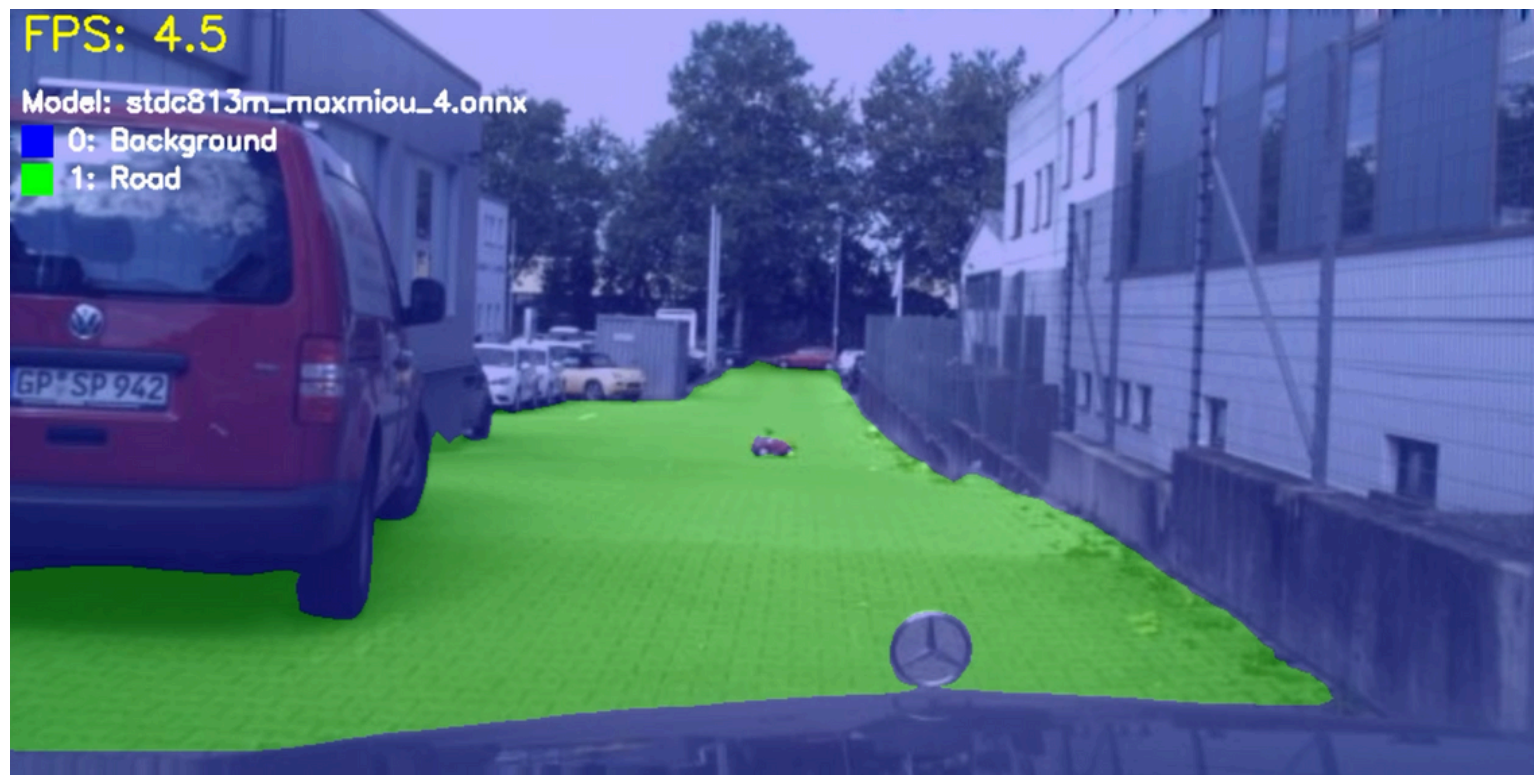
3 classes : road, obstacles and background

iterations : 50000

mIOU50: 0.66

Conclusion : unusable

Road Segmentation v4 : new config, bigger dataset



Dataset : cityscapes
2 classes only : road and background
iterations : 50000
mIOU50: 0.95
Conclusion : Good

Object detection



First iteration



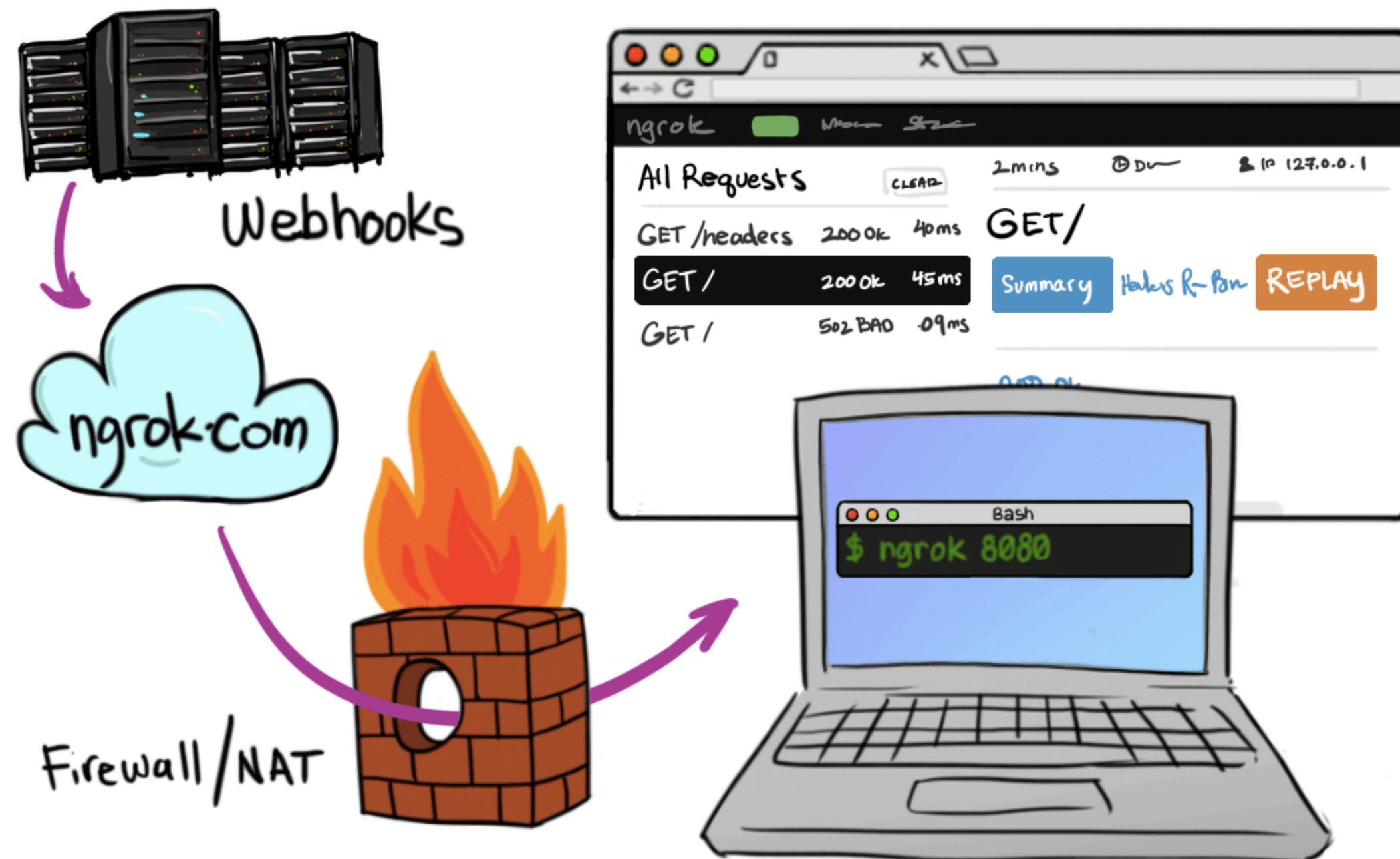
Second iteration

Dataset : imad's dataset → imad's dataset (balanced) - speed signs + lost and found
1 class: obstacles

P : 0.8, R : 0.71, mAP50 : 0.76 → P : 0.77, R : 0.64, mAP50 : 0.68

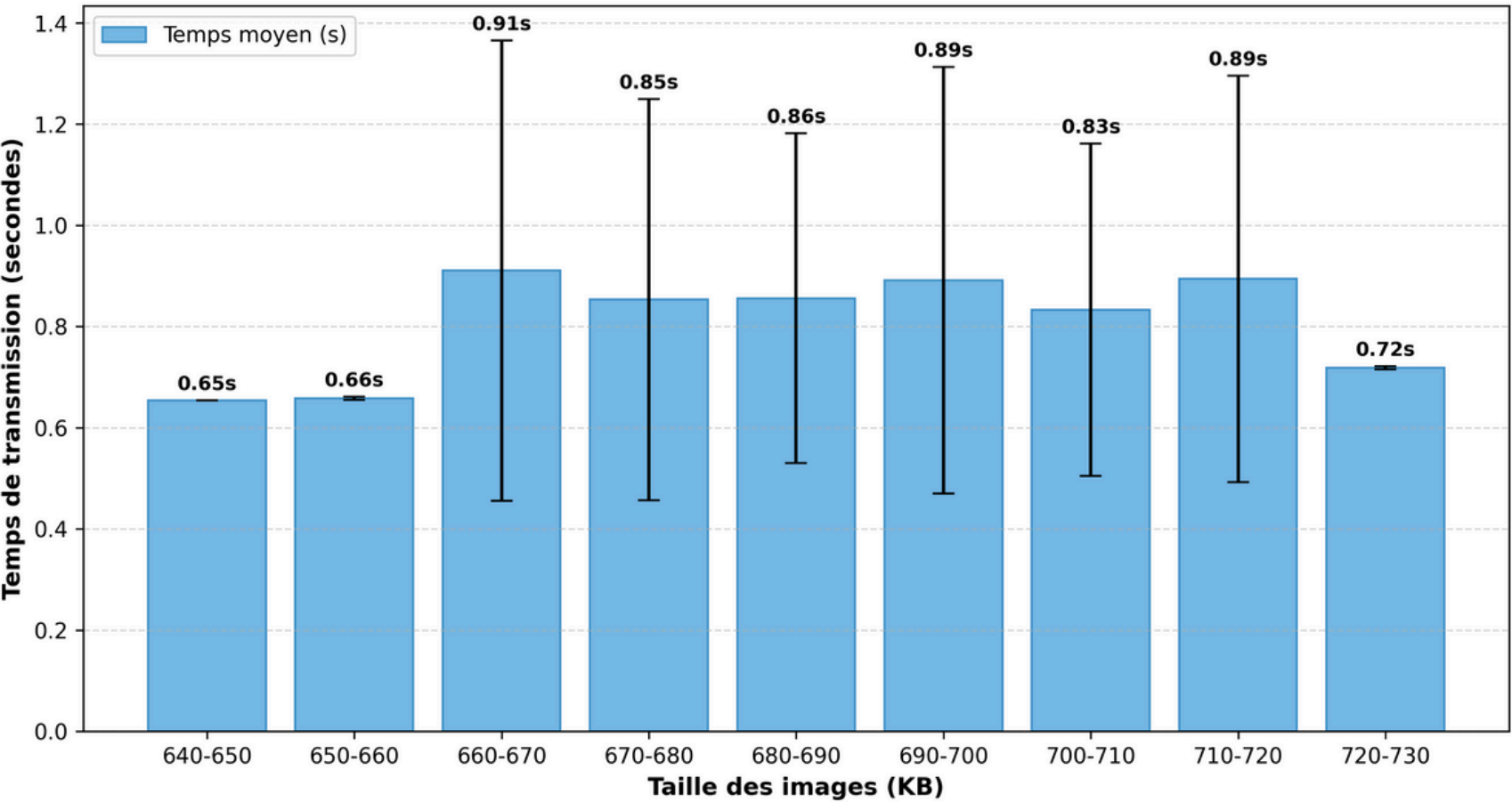
Connectivité Raspberry Pi-to-GPU & Tunneling

Connectivité Raspberry Pi-to-GPU avec ngrok

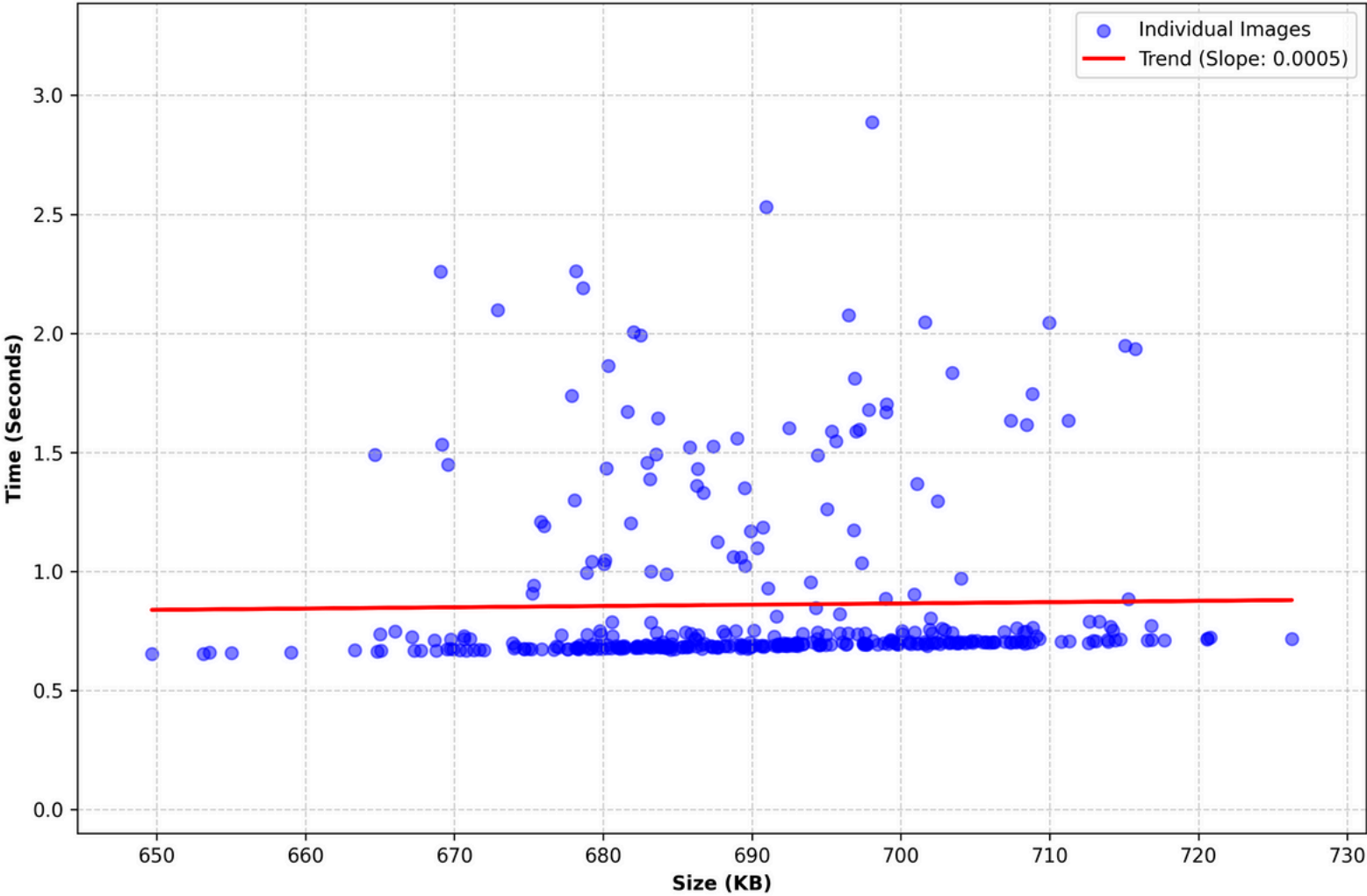


Benchmark de Performance

Benchmark Performance : Latence de transmission Edge-to-GPU
(Moyenne et Écart-Type)

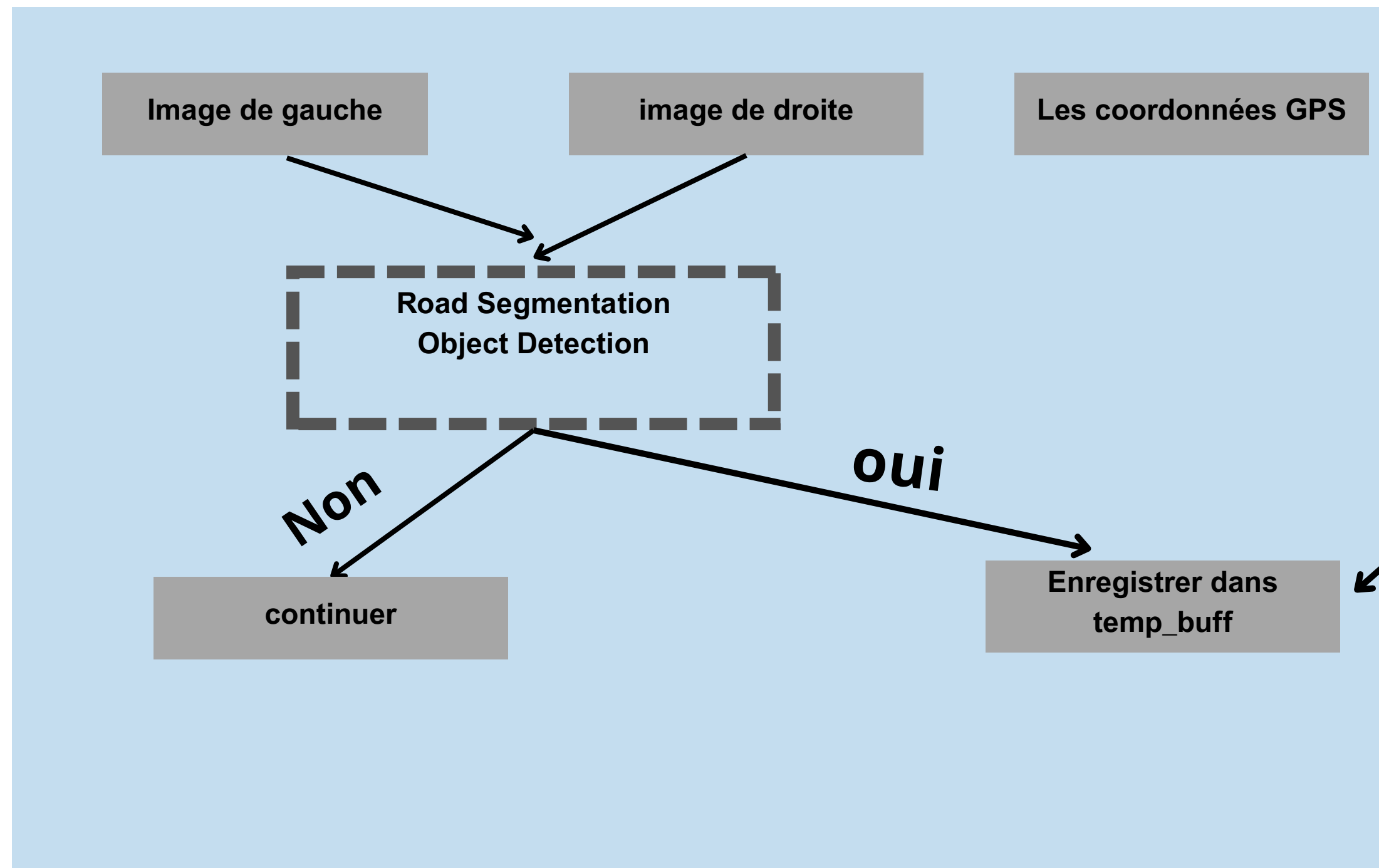


Impact of Image Size on Transmission Time



Vitesse moyenne WiFi : 3,94 Mbps

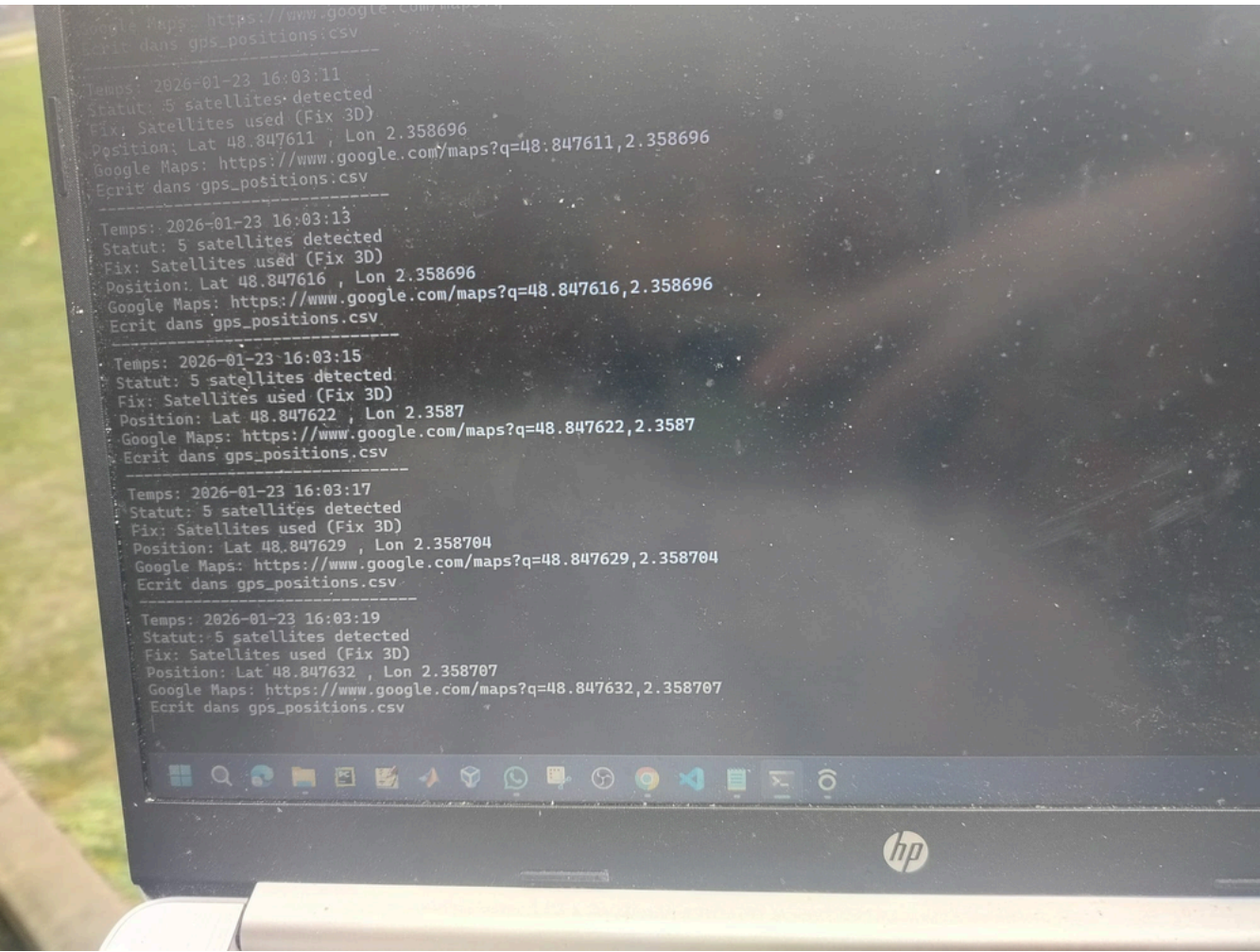
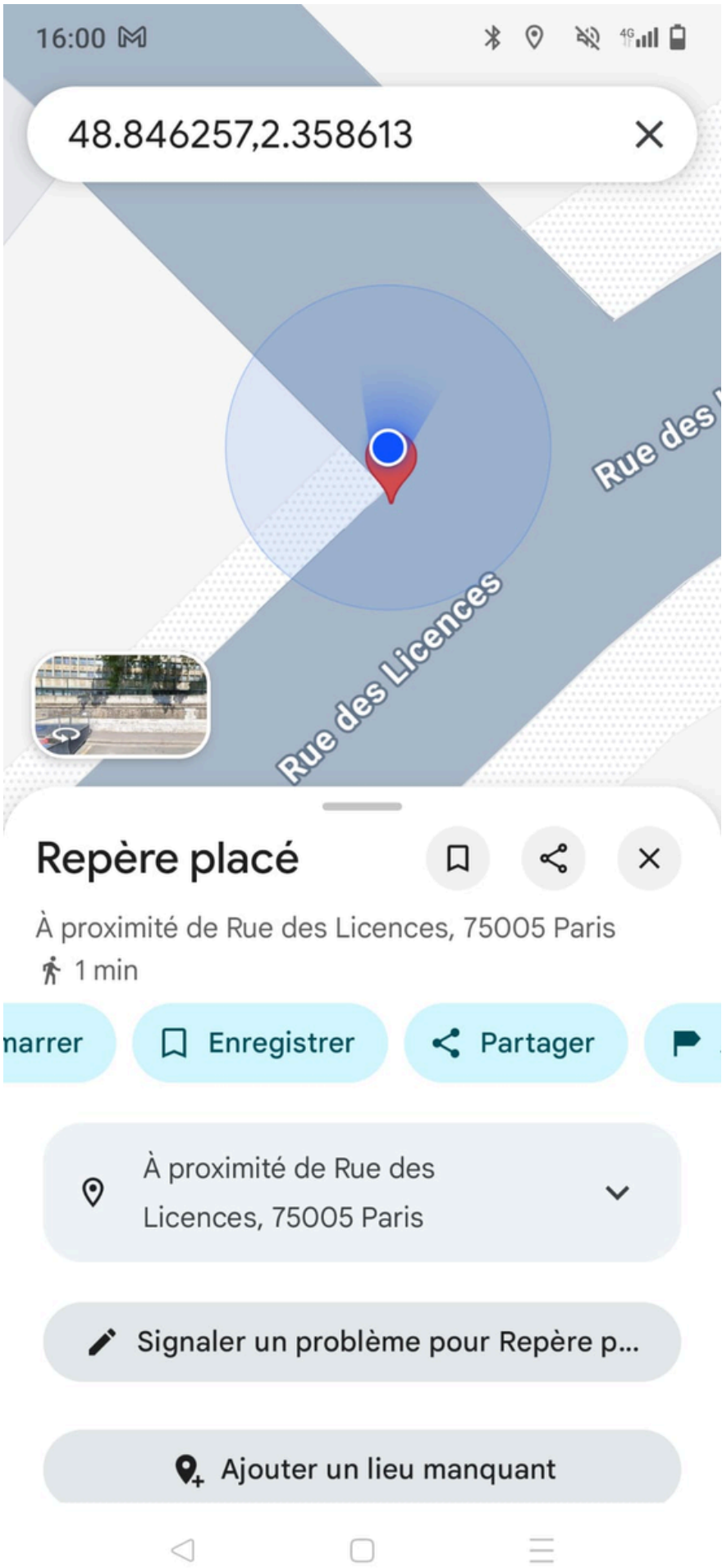
Le code principal est dans la Raspberry Pi.



Le code sender_worker.py.

Vérifier s'il y a des données dans temp_buff, les envoyer, puis les supprimer.

Intégration et Validation du Module GPS



Point	GPS		Google Maps		Erreur (°)		Erreur (m)	
	Latitude	Longitude	Latitude	Longitude	Δ Lat	Δ Lon	Δ Lat	Δ Lon
1	48.847405	2.358092	48.846753	2.357957	0.000652	0.000135	72.4	9.9
2	48.847271	2.357733	48.847060	2.357947	0.000211	0.000214	23.4	15.6
3	48.847263	2.358916	48.846943	2.358969	0.000320	0.000053	35.5	3.9
4	48.846019	2.357775	48.845816	2.357751	0.000203	0.000024	22.5	1.8
5	48.846619	2.359084	48.846257	2.358613	0.000362	0.000471	40.2	34.4

Classification des obstacles

Datasets utilisés, objectifs et classes

Dataset	Objectif	Classes détectées
LostAndFound	Obstacles inhabituels	Objets non standards sur la chaussée
Road Works Sign	Zones de travaux	Panneau lumineux (flèche), cônes de signalisation
Speed Bumps	Événements routiers	Ralentisseur
Stop Sign	Signalisation réglementaire	Panneau Stop
Road Damage	Dégradations de chaussée	Nid-de-poule, fissures en réseau, fissures latérales, fissures longitudinales
Road Debris 1	Débris sur la route	Débris et objets divers sur la chaussée
Road Obstacle	Obstacles routiers	Déchets, bloc routier, arbre tombé, etc.

Métriques d'évaluation (YOLOv8)

Les performances sont évaluées à l'aide des métriques standard YOLO : précision (P), rappel (R), mAP@50 et mAP@50–95

Modèle	Précision (P)	Rappel (R)	mAP@50	mAP@50–95
Obstacle_LAF	0.942	0.962	0.982	0.732
RoadWorks	0.982	0.984	0.989	0.964
SpeedBumps	0.938	0.874	0.906	0.541
StopSign	0.977	0.937	0.982	0.861
RoadDamage	0.765	0.709	0.759	0.450
RoadDebris1	0.663	0.621	0.588	0.432
RoadObstacle	0.798	0.738	0.776	0.425

Métriques d'évaluation (YOLOv8 Modèle unique)

mAP@50-95	mAP@50	Precision	Recall
0.653	0.866	0.878	0.829

TABLE 1 – Métriques de performance du meilleur modèle

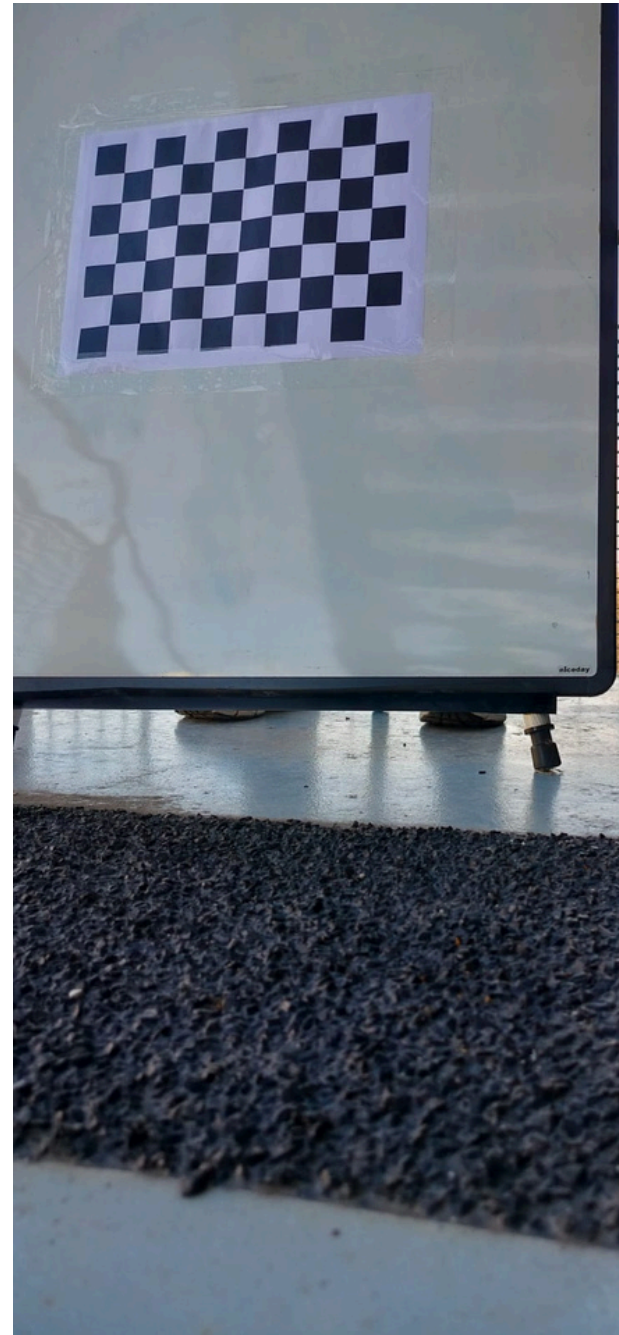
Prétraitement (ms)	Inférence (ms)	Post-traitement (ms)
0.90	13.61	1.82

TABLE 2 – Temps d'exécution du meilleur modèle

Résultats Subjectifs un peu moins fiables !

Estimation de volume

stereo Calibration



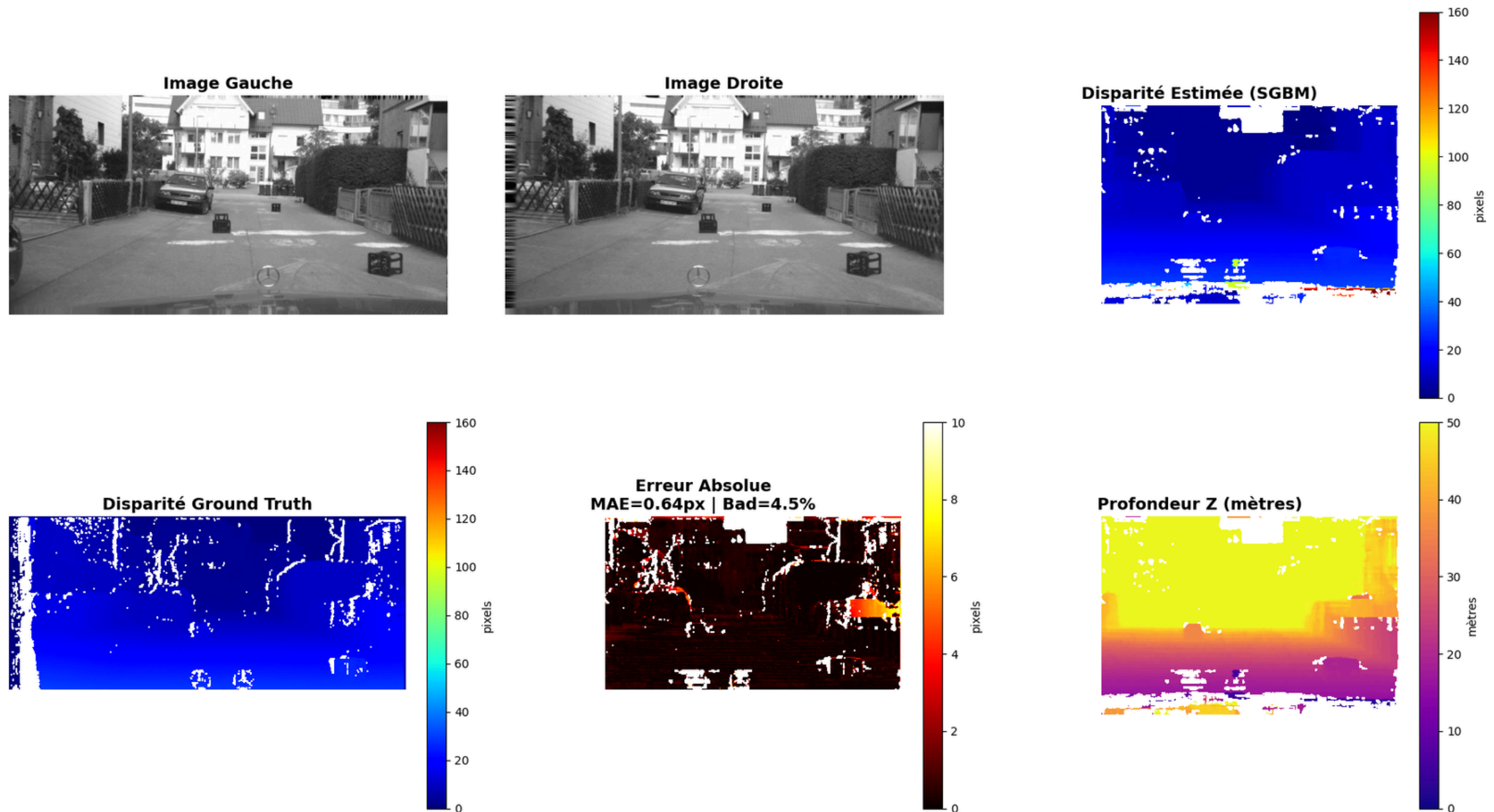
$$Z = \frac{f \cdot B}{d}$$

- Z : Distance réelle de l'objet (mètres).
- f : Focale de la caméra (pixels).
- B : Écart entre les deux caméras (mètres).
- d : Décalage de l'objet entre l'image gauche et droite (pixels).

Vecteur T (en mm) :

```
[ [-799.2311519 ]  
[ -12.81957111]  
[ 171.64444609]]
```


Calcul de la disparité



Estimation de volume



Resultat:0.309m²@260.06m

Application mobile